# Lecture 15

## Perturbation Theory for Linear Operators

This optional capstone asks what remains when an operator cannot be diagonalized exactly but is close to one that can. Starting from the spectral decomposition of Lecture 10 and the norm, spectrum, and resolvent language of Lecture 14, we study self-adjoint families

$$H(\varepsilon) = H_0 + \varepsilon V.$$

The spectral gap controls both what can be proved and what can be approximated. No core lecture depends on this material, and it is outside the mandatory assessment unless an instructor explicitly includes it.

**Remark 15.1 (Optional-capstone scope).** The first two main sections carry out the perturbative derivations on a finite-dimensional complex inner-product space. The third section, "Beyond finite dimensions", changes scope explicitly to complex Hilbert spaces, uses $\mathcal{B}(\mathcal{H})$ for bounded operators, and treats an unbounded operator together with its domain. This lecture is optional; the 14-lecture core ends at Lecture 14.

**Learning objectives**

By the end of this lecture you will be able to:

- Use the operator norm and resolvent to bound how far a spectrum can move.
- Compute first- and second-order shifts of a simple eigenvalue and the first-order change of its eigenvector.
- State and apply the Hellmann–Feynman identity.
- Diagnose the failure of non-degenerate formulas and obtain the first-order branch slopes from $PVP$, including the case of residual degeneracy.
- **Extension/outlook:** explain which perturbative ideas survive, and which can fail, in infinite dimensions.

## 1 Recommended optional route: spectral size and the resolvent

Lecture 14 defined the operator norm and resolvent for bounded operators. In the finite-dimensional normal case, Lecture 10's spectral projections make both quantities completely explicit.

One more quantity is read straight off the spectrum. How much can an operator stretch a vector?

**Definition 15.2 (Operator norm).** For $T \in \mathcal{L}(V)$, the *operator norm* is

$$\|T\| := \sup_{v\neq 0} \frac{\|Tv\|}{\|v\|} = \sup_{\|v\|=1} \|Tv\|.$$

In the present finite-dimensional setting the unit sphere is compact, so the continuous function $v \mapsto \|Tv\|$ attains this supremum and it is a maximum. At once $\|Tv\| \leq \|T\|\,\|v\|$, and hence $\|ST\| \leq \|S\|\,\|T\|$.

**Proposition 15.3 (Norm and resolvent of a normal operator).** Let $T = \sum_k \lambda_k P_k$ be normal on the finite-dimensional complex space $V$. Then

$$\|T\| = \max_k |\lambda_k|,$$

and for every $z \notin \operatorname{spec}(T)$ the operator $T - zI$ is invertible with

$$\|(T - zI)^{-1}\| = \frac{1}{\operatorname{dist}\big(z, \operatorname{spec}(T)\big)}, \qquad \operatorname{dist}\big(z, \operatorname{spec}(T)\big) = \min_k |\lambda_k - z|.$$

*Proof.* Resolve $v = \sum_k P_k v$ into orthogonal eigen-components. By Pythagoras (Lecture 7), $\|v\|^2 = \sum_k \|P_k v\|^2$ and

$$\|Tv\|^2 = \sum_k |\lambda_k|^2 \|P_k v\|^2 \le \Big(\max_k |\lambda_k|\Big)^2 \|v\|^2,$$

with equality when $v$ is an eigenvector for a largest-modulus eigenvalue; hence $\|T\| = \max_k |\lambda_k|$. For the second claim, the functional calculus applied to $f(\lambda) = (\lambda - z)^{-1}$ gives the normal operator $(T - zI)^{-1} = \sum_k (\lambda_k - z)^{-1} P_k$, whose norm is therefore $\max_k |\lambda_k - z|^{-1} = \big(\min_k |\lambda_k - z|\big)^{-1}$. ■

The resolvent $(T - zI)^{-1}$ thus blows up exactly as $z$ approaches the spectrum, at a rate set by nothing but the distance to the nearest eigenvalue.

## 2 Recommended optional route: perturbing a spectral decomposition

Exact diagonalization is the exception, not the rule. Almost every operator a physicist actually meets has the shape $H(\varepsilon) = H_0 + \varepsilon V$, with $H_0$ self-adjoint and already diagonalized, $V$ self-adjoint, and $\varepsilon \in \mathbb{R}$ small: a spin in a slightly tilted field, an atom in a weak electric field, a lattice with one defective site. Two questions, in this order. How far can the spectrum move at all? And, knowing it moves only a little, what is the shift to leading orders in $\varepsilon$? Throughout, a denominator is written with the level of interest first, $E_n - E_m$.

### How far can the spectrum move?

The first question has a clean answer, with Proposition 15.3 as the engine. It needs one standard tool, the operator version of $\frac{1}{1-x} = 1 + x + x^2 + \cdots$.

**Lemma 15.4 (Neumann series).** If $X \in \mathcal{L}(V)$ satisfies $\|X\| < 1$, then $I - X$ is invertible and

$$(I - X)^{-1} = \sum_{n \ge 0} X^n, \qquad \|(I - X)^{-1}\| \le \frac{1}{1 - \|X\|}.$$

*Proof.* Sub-multiplicativity (Definition 15.2) gives $\|X^n\| \le (\|X\|)^n$, so the series converges absolutely in the finite-dimensional space $\mathcal{L}(V)$. Its partial sums satisfy $(I - X)S_N = S_N(I - X) = I - X^{N+1} \to I$, since $\|X^{N+1}\| \le (\|X\|)^{N+1} \to 0$; hence the limit $S$ obeys $(I - X)S = S(I - X) = I$, with $\|S\| \le \sum_n \|X\|^n = (1 - \|X\|)^{-1}$. ■

**Theorem 15.5 (Spectral stability).** Let $H_0$ and $V$ be self-adjoint, $\varepsilon \in \mathbb{R}$, and let $E_1, \ldots, E_r$

be the distinct eigenvalues of $H_0$. Then

$$\operatorname{spec}\big(H_0 + \varepsilon V\big) \subseteq \bigcup_{k=1}^{r} \big\{ z \in \mathbb{C} : |z - E_k| \le |\varepsilon|\, \|V\| \big\}.$$

*Proof.* Let $z$ satisfy $\operatorname{dist}(z, \operatorname{spec} H_0) > |\varepsilon|\|V\|$; we show $H_0 + \varepsilon V - zI$ is invertible. First, $z \notin \operatorname{spec}(H_0)$, so $H_0 - zI$ is invertible, and we may factor

$$H_0 + \varepsilon V - zI = (H_0 - zI)\big(I + \varepsilon (H_0 - zI)^{-1} V\big).$$

Since $H_0$ is normal, Proposition 15.3 and sub-multiplicativity give

$$\big\| -\varepsilon (H_0 - zI)^{-1} V \big\| \le |\varepsilon|\, \|(H_0 - zI)^{-1}\|\, \|V\| = \frac{|\varepsilon|\, \|V\|}{\operatorname{dist}(z, \operatorname{spec} H_0)} < 1,$$

so the second factor is invertible by Lemma 15.4. A product of invertible operators is invertible, so $z \notin \operatorname{spec}(H_0 + \varepsilon V)$. ■

**Remark 15.6 (The Neumann–stability proof spine).** The argument has no hidden algebra. Submultiplicativity gives $\|X^n\| \le (\|X\|)^n$, so completeness supplies the norm limit of the geometric series. Its partial sums obey both identities $(I - X)S_N = S_N(I - X) = I - X^{N+1}$, proving a two-sided inverse and the bound $\|(I - X)^{-1}\| \le (1 - \|X\|)^{-1}$. For spectral stability, set $R_0(z) = (H_0 - zI)^{-1}$ and factor

$$H_0 + \varepsilon V - zI = (H_0 - zI)(I + \varepsilon R_0(z) V).$$

The normal resolvent formula makes the second factor Neumann-invertible whenever $\operatorname{dist}(z, \operatorname{spec} H_0) > |\varepsilon|\|V\|$. Swapping the two self-adjoint operators supplies the reverse spectral inclusion.

Every perturbed eigenvalue therefore stays within $|\varepsilon|\|V\|$ of an unperturbed one. Applying the same theorem with $H_0 + \varepsilon V$ as the reference operator and perturbation $-\varepsilon V$ gives the reverse inclusion, so the two spectra are Hausdorff-close by at most $|\varepsilon|\|V\|$. Thus the spectrum is *Lipschitz* in the perturbation—no smallness assumption, no series. The work was done by the resolvent bound of Proposition 15.3, which is exactly where self-adjointness entered.

**Proposition 15.7 (No level is lost · cited).** If $2|\varepsilon|\|V\| < \min_{j \ne k} |E_j - E_k|$, the disks of Theorem 15.5 are pairwise disjoint, and each contains exactly as many eigenvalues of $H_0 + \varepsilon V$, with multiplicity, as $H_0$ has at its centre.

Using this cited level-count statement belongs to the recommended optional route; proving it via continuity of spectral projections is an optional extension.

A small perturbation thus moves levels without destroying them or letting two groups merge, provided the disks stay apart. The controlling ratio is *perturbation size over spectral gap*—a theme that returns in every formula below; Example 15.15 works it out.

### First- and second-order corrections

Now assume every eigenvalue of $H_0$ is *simple* (non-degenerate), with $H_0|m\rangle = E_m|m\rangle$ for an orthonormal basis $\{|m\rangle\}$ and eigenprojections $P_m = |m\rangle\langle m|$. Fix a level $|n\rangle$.

**Theorem 15.8 (Analyticity of a simple level · sketched).** If $E_n$ is a simple eigenvalue of $H_0$, then for small $|\varepsilon|$ the operator $H(\varepsilon)$ has exactly one eigenvalue $E_n(\varepsilon)$ near $E_n$, and it—with a suitable choice of eigenvector—is given by a convergent power series in $\varepsilon$. Here the real physical parameter is complexified in a neighbourhood of zero solely to make the analytic statement.

*Sketch.* Uniqueness is Proposition 15.7. For the series, apply the implicit function theorem to $p(\lambda, \varepsilon) = \det(\lambda I - H(\varepsilon))$, a polynomial in both arguments: $p(E_n, 0) = 0$ and $\partial p/\partial\lambda \neq 0$ there, *precisely because* $E_n$ is a simple root, giving a unique analytic branch through $E_n$. That settles the eigenvalue but says nothing about the eigenvector, so apply the same theorem a second time, to the pair

$$F(v, \lambda, \varepsilon) = \big(H(\varepsilon)v - \lambda v,\ \langle n \mid v\rangle - 1\big),$$

which vanishes at $(|n\rangle, E_n, 0)$. Its derivative there in $(v, \lambda)$ sends $(w, \mu)$ to $\big((H_0 - E_n)w - \mu|n\rangle,\ \langle n \mid w\rangle\big)$; pairing the first component with $\langle n|$ kills the left-hand side and forces $\mu = 0$, after which simplicity of $E_n$ gives $w \in \operatorname{span}(|n\rangle)$ and $\langle n \mid w\rangle = 0$ gives $w = 0$. The derivative is therefore injective. Its finite-dimensional domain and codomain have the same dimension, so it is invertible, and $|n(\varepsilon)\rangle$ is analytic as well—in the intermediate normalization used in (15.1). ■

**Remark 15.9 (Analytic licence and intermediate normalization).** The determinant argument is finite-dimensional and complex: a simple root has $\partial_\lambda \det(\lambda I - H(\varepsilon)) \neq 0$, so the complex implicit function theorem gives one local analytic eigenvalue branch. Applying it to the eigenpair equation with gauge $\langle n \mid n(\varepsilon)\rangle = 1$ gives an analytic eigenvector branch because the displayed derivative in the proof is injective and hence invertible between equal finite dimensions. This gauge is *intermediate normalization*; it fixes scale but generally gives $\|n(\varepsilon)\| = 1 + O(\varepsilon^2)$, not a unit vector.

This licenses the ansatz; without it what follows would be formal. Write

$$E_n(\varepsilon) = E_n + \varepsilon E_n^{(1)} + \varepsilon^2 E_n^{(2)} + \cdots, \qquad |n(\varepsilon)\rangle = |n\rangle + \varepsilon|n^{(1)}\rangle + \varepsilon^2|n^{(2)}\rangle + \cdots, \tag{15.1}$$

using *intermediate normalization* $\langle n \mid n(\varepsilon)\rangle = 1$, i.e. $\langle n \mid n^{(k)}\rangle = 0$ for $k \geq 1$ (the eigenvector is fixed only up to scale; this is the convenient choice, not unit normalization). Substituting into $H(\varepsilon)|n(\varepsilon)\rangle = E_n(\varepsilon)|n(\varepsilon)\rangle$, the terms of order $\varepsilon$ and $\varepsilon^2$ read

$$(H_0 - E_n)|n^{(1)}\rangle = \big(E_n^{(1)} - V\big)|n\rangle, \qquad (H_0 - E_n)|n^{(2)}\rangle = \big(E_n^{(1)} - V\big)|n^{(1)}\rangle + E_n^{(2)}|n\rangle. \tag{15.2}$$

Pair the first with $\langle n|$: the left side vanishes, since $H_0$ is self-adjoint with $E_n$ real, so $\langle n|\,(H_0 - E_n)\,|n^{(1)}\rangle = \langle (H_0 - E_n)n, n^{(1)}\rangle = 0$, while the right side is $E_n^{(1)} - \langle n| V |n\rangle$. Pairing it with $\langle m|$ for $m \neq n$ gives $(E_m - E_n)\langle m \mid n^{(1)}\rangle = -\langle m| V |n\rangle$. Pairing the second with $\langle n|$ kills its left side for the same reason, leaving $0 = -\langle n| V |n^{(1)}\rangle + E_n^{(2)}$. Together:

**Proposition 15.10 (Rayleigh–Schrödinger corrections).** Under the simple-level and intermediate-normalization hypotheses above,

$$\boxed{E_n^{(1)} = \langle n| V |n\rangle, \qquad |n^{(1)}\rangle = \sum_{m\neq n} \frac{\langle m| V |n\rangle}{E_n - E_m}|m\rangle, \qquad E_n^{(2)} = \sum_{m\neq n} \frac{|\langle m| V |n\rangle|^2}{E_n - E_m}} \tag{15.3}$$

(the last using $\langle n| V |m\rangle = \overline{\langle m| V |n\rangle}$, since $V$ is self-adjoint). These are the *Rayleigh–Schrödinger* formulas: a level shifts by the diagonal matrix element of $V$ in its own state, the state picks up an admixture of every other state weighted by coupling over gap, and at second order the level shifts by a sum over all other levels, coupling squared over gap.

**Remark 15.11 (The finite reduced resolvent).** Formula (15.3) looks basis-dependent but is not. With the *reduced resolvent* $S_n := \sum_{m\neq n} P_m/(E_n - E_m)$—the inverse of $E_n - H_0$ on the orthogonal complement of $|n\rangle$—it reads

$$|n^{(1)}\rangle = S_n V|n\rangle, \qquad E_n^{(2)} = \langle n|\, V S_n V\,|n\rangle,$$

written entirely in the eigenprojections from the spectral-decomposition theorem of Lecture 10. This is the form that extends beyond finite dimensions, where $S_n$ is built from the resolvent $(H_0 - z)^{-1}$ rather than from a finite sum.

**Remark 15.12 (Level repulsion, and two free checks).** Every denominator in $E_n^{(2)}$ is negative for the *lowest* level and positive for the highest, so the ground-state correction is non-positive and the top-state correction is non-negative; either may be zero when all relevant couplings vanish. Two identities check any calculation—summing the first of (15.3) gives $\sum_n E_n^{(1)} = \operatorname{tr} V$, and pairing the $(m,n)$ and $(n,m)$ terms of the third, whose denominators differ only in sign, gives $\sum_n E_n^{(2)} = 0$. Both are forced by $\operatorname{tr} H(\varepsilon) = \operatorname{tr} H_0 + \varepsilon \operatorname{tr} V$ being exactly linear in $\varepsilon$.

**Remark 15.13 (When is the perturbation small?).** The mixing magnitude is not controlled by $\varepsilon$ alone but by the dimensionless ratio $|\varepsilon||\langle m|\, V\,|n\rangle|/|E_n - E_m|$: coupling measured against the *gap*. A nominally tiny $\varepsilon$ buys nothing when two levels are nearly degenerate—the situation the next subsection addresses.

**Theorem 15.14 (Hellmann–Feynman).** Let $H(\varepsilon) = H_0 + \varepsilon V$ and let $H(\varepsilon)|n(\varepsilon)\rangle = E_n(\varepsilon)|n(\varepsilon)\rangle$, with a differentiable nonzero eigenvector branch. Then

$$\frac{dE_n}{d\varepsilon} = \frac{\langle n(\varepsilon)|\, V\,|n(\varepsilon)\rangle}{\langle n(\varepsilon)\,|\, n(\varepsilon)\rangle}.$$

For a unit eigenvector the denominator is one. For the intermediate-normalized branch of (15.1), the norm is generally $1 + O(\varepsilon^2)$ rather than exactly one.

*Proof.* Differentiate the eigenvalue equation, pair it with $\langle n(\varepsilon)|$, and use self-adjointness:

$$\langle n|\, V\,|n\rangle + \langle n|\, H\,|\dot n\rangle = \dot E_n\langle n\,|\, n\rangle + E_n\langle n\,|\,\dot n\rangle, \qquad \langle n|\, H\,|\dot n\rangle = E_n\langle n\,|\,\dot n\rangle.$$

The state-derivative terms cancel, leaving the displayed identity. At $\varepsilon = 0$ it gives $E_n^{(1)} = \langle n|\, V\,|n\rangle$. Exercise 15.5 asks you to reconstruct the equivalent unit-normalized Rayleigh-quotient proof. ■

**Example 15.15 (A three-level system, worked out).** Take, in the standard basis $|1\rangle, |2\rangle, |3\rangle$,

$$H_0 = \operatorname{diag}(0,1,3), \qquad V = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -3 \end{pmatrix}.$$

First, stability. The eigenvalues of $V$ are $-3.3460,\ 0.4067,\ 2.9392$, so $\|V\| = 3.3460$ by Proposition 15.3, while the smallest gap of $H_0$ is 1. Proposition 15.7 therefore applies while $2|\varepsilon|(3.3460) < 1$, i.e. $|\varepsilon| < 0.149$: up to there the three levels sit in three disjoint disks and can be tracked one by one.

Now the shifts. To first order they are the diagonal entries of $V$,

$$E_1^{(1)} = 2, \qquad E_2^{(1)} = 1, \qquad E_3^{(1)} = -3,$$

summing to $0 = \operatorname{tr} V$ as they must. All off-diagonal entries equal 1, so the second-order shifts are

$$E_1^{(2)} = \frac{1}{0-1} + \frac{1}{0-3} = -\frac{4}{3}, \qquad E_2^{(2)} = \frac{1}{1-0} + \frac{1}{1-3} = \frac{1}{2}, \qquad E_3^{(2)} = \frac{1}{3-0} + \frac{1}{3-1} = \frac{5}{6},$$

summing to $-\frac{4}{3} + \frac{1}{2} + \frac{5}{6} = 0$, the second check. Read physically, the lowest level is pushed *down* by both partners, the highest *up* by both, and the middle one, with a partner on each side, barely moves. For the ground state $E_1(\varepsilon) \approx 2\varepsilon - \frac{4}{3}\varepsilon^2$, against the exact eigenvalue:

| $\varepsilon$ | first order | second order | exact |
|---|---|---|---|
| 0.05 | 0.1 | 0.096667 | 0.096562 |
| 0.10 | 0.2 | 0.186667 | 0.185892 |
| 0.20 | 0.4 | 0.346667 | 0.341699 |

Doubling $\varepsilon$ multiplies the first-order error by about 4 and the second-order error by about 8—the errors are $O(\varepsilon^2)$ and $O(\varepsilon^3)$, as (15.1) predicts—and the accuracy visibly deteriorates at $\varepsilon = 0.2$, just past the 0.149 threshold beyond which the disjoint-disk guarantee no longer applies. The first-order *state* is equally concrete: (15.3) gives $|1(\varepsilon)\rangle \approx (1, -0.1, -0.0333)$ at $\varepsilon = 0.1$. Scaled to the same intermediate normalization $\langle 1 \,|\, 1(\varepsilon)\rangle = 1$, the exact eigenvector is $(1, -0.1055, -0.0356)$—the leading component agrees identically, by the normalization, and the two admixtures are right to within about 6%. (Its unit-normalized form is $(0.9939, -0.1049, -0.0354)$; comparing against *that* would charge the approximation for a change of normalization.) ◁

## Degenerate levels

Formula (15.3) divides by zero the moment two levels coincide, and Theorem 15.8 does not apply, having needed a *simple* root. This is no technical blemish: a degenerate level may have several first-order branches, but those branches need not have distinct slopes. The resolution is a reading of the degenerate-observable example from Lecture 10, where a degenerate eigenvalue was seen to leave freedom in the eigenbasis. The perturbation may reduce that freedom, but it need not remove it completely.

> **Theorem 15.16 (First-order branches at a degenerate level • sketched).** Let $E$ be an eigenvalue of $H_0$ of finite multiplicity $d$, with spectral projection $P$, and put $Q = I - P$. The multiset of first-order branch slopes, counted with multiplicity, is the spectrum of the finite self-adjoint compression $PVP|_{\operatorname{range} P}$. If those slopes are distinct, $PVP$ resolves the degeneracy and its eigenvectors determine the zeroth-order branch states up to phase. If a slope is repeated, its eigenspace remains degenerate at first order; no basis inside that repeated eigenspace is selected, and a higher-order effective operator may be required.

*Sketch.* Granting that the perturbed eigenvalue and eigenvector admit expansions (15.1) about some $|\psi\rangle \in \operatorname{range} P$ (for a self-adjoint family this is Rellich's theorem, quoted here), the order-$\varepsilon$ equation reads $(H_0 - E)|\psi^{(1)}\rangle = \big(E^{(1)} - V\big)|\psi\rangle$. Apply $P$: since $H_0$ commutes with $P$ and $(H_0 - E)P = 0$, the left side vanishes, leaving $0 = E^{(1)}|\psi\rangle - PVP|\psi\rangle$ (using $P|\psi\rangle = |\psi\rangle$). So $|\psi\rangle$ is an eigenvector of $PVP$ on range $P$ with eigenvalue $E^{(1)}$; and $PVP$ being self-adjoint on the $d$-dimensional range $P$, the spectral theorem supplies $d$ orthonormal such states with real eigenvalues. ■

The compression therefore determines the first-order slopes, with multiplicity; it produces a splitting and a preferred basis only to the extent that its own eigenvalues are distinct. For the inoculating counterexample $H_0 = \operatorname{diag}(1, 1, 4)$ and $V = I$, one has $PVP = P$: both branches have slope 1, there is no first-order splitting, and every orthonormal basis of range $P$ is equally valid. Zeeman and Stark splitting are important cases where the compression does have distinct slopes, not a universal outcome.

**Example 15.17 (A degeneracy split).** Take $H_0 = \mathrm{diag}(1,1,4)$ and $V$ with rows $(0,2,1)$, $(2,0,1)$, $(1,1,0)$. The non-degenerate formula is meaningless here: $\langle 1|V|2\rangle = 2 \neq 0$ while $E_1 - E_2 = 0$. Instead compress $V$ onto range $P = \mathrm{span}(|1\rangle, |2\rangle)$:

$$PVP\big|_{\mathrm{range}\,P} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = 2\sigma_x, \qquad \mu = \pm 2,$$

so this degenerate level splits as $E \approx 1 \pm 2\varepsilon$, with zeroth-order branch states $|\pm\rangle = \frac{1}{\sqrt{2}}(|1\rangle \pm |2\rangle)$, determined up to phase because the slopes are distinct—the $\sigma_x$ eigenstates from the Pauli-$\sigma_x$ example in Lecture 10. At $\varepsilon = 0.1$ the exact eigenvalues are $0.800000,\ 1.192875,\ 4.007125$ against the predicted $0.8,\ 1.2,\ 4$.

Two things are worth noticing. The lower branch $1 - 2\varepsilon$ is *exact to all orders*, for a reason worth more than any series: $\langle 3|V|-\rangle = 0$, so $|-\rangle$ is an exact eigenvector of $H(\varepsilon)$ for every $\varepsilon$. For second order one must still respect the original degenerate eigenspace. With $Q = I - P$ and $S = Q[(1 - H_0)|_{\mathrm{range}\,Q}]^{-1}Q$, the complementary-space formula gives

$$E_+^{(2)} = \langle +|VSV|+\rangle = \frac{|\langle 3|V|+\rangle|^2}{1-4} = \frac{|\sqrt{2}|^2}{-3} = -\tfrac{2}{3}.$$

Thus $1 + 2\varepsilon - \frac{2}{3}\varepsilon^2 = 1.193333$ at $\varepsilon = 0.1$, against the exact $1.192875$. ◁

## 3 Optional extension/outlook: beyond finite dimensions

**Remark 15.18 (Optional extension: bounded-perturbation resolvent expansion).** From this point $\mathcal{H}$ is a complex Hilbert space and $\mathcal{B}(\mathcal{H})$ denotes the bounded operators. For the resolvent expansion, let $H_0$ be self-adjoint, possibly unbounded, on $D(H_0)$; let $V \in \mathcal{B}(\mathcal{H})$ be bounded and self-adjoint; and take $z \in \rho(H_0)$. Then $R_0(z) = (H_0 - z\,\mathrm{id})^{-1}$ is bounded from $\mathcal{H}$ into $D(H_0)$, and on $D(H_0)$ one has the factorization

$$H_0 + \varepsilon V - z\,\mathrm{id} = \big(I + \varepsilon V R_0(z)\big)(H_0 - z\,\mathrm{id}).$$

Consequently, if $|\varepsilon|\,\|VR_0(z)\| < 1$, Neumann inversion gives

$$\big(H_0 + \varepsilon V - z\,\mathrm{id}\big)^{-1} = \sum_{n\ge 0} (-\varepsilon)^n R_0(z)\big(VR_0(z)\big)^n,$$

with convergence in operator norm. In finite dimensions its poles track eigenvalues. In Hilbert space the spectrum may also be continuous, and later unbounded perturbations require separate domain hypotheses.

**Proposition 15.19 (Optional extension: reduced resolvent for an isolated eigenvalue).** Let $E$ be an isolated eigenvalue of a self-adjoint $H_0$ on $D(H_0)$, and let $\mathsf{E}_{H_0}$ be its PVM. Set $P_E = \mathsf{E}_{H_0}(\{E\})$ and $Q = I - P_E$. Isolation supplies a gap $\delta = \mathrm{dist}(E, \mathrm{spec}(H_0) \setminus \{E\}) > 0$. The restriction of $E - H_0$ has domain $D(H_0) \cap Q\mathcal{H}$ and maps it bijectively onto $Q\mathcal{H}$. Its bounded inverse, extended by zero on $P_E\mathcal{H}$, is the *reduced resolvent*

$$S = Q\big[(E - H_0)|_{D(H_0)\cap Q\mathcal{H}}\big]^{-1}Q = \int_{\mathrm{spec}(H_0)\setminus\{E\}} \frac{1}{E-\lambda}\, d\mathsf{E}_{H_0}(\lambda), \qquad \|S\| \le \delta^{-1}.$$

The singular atom at $E$ is excluded. If $E$ is embedded or non-isolated, this bounded reduced inverse need not exist. For a finite spectral list, $S = \sum_{E_m \neq E} P_m/(E - E_m)$, which

> becomes the $m \neq n$ formula for a simple level.

**Remark 15.20 (Optional extension: perturbation theory in infinite dimensions).** The finite-dimensional stability argument and the bounded-perturbation resolvent series transfer under the hypotheses just stated. Spectral stability survives unchanged for bounded self-adjoint operators, because its only ingredient—that $\|(T - z\,\mathrm{id})^{-1}\| = 1/\mathrm{dist}(z, \operatorname{spec} T)$ for normal $T$—is a consequence of the spectral theorem, not of finite dimensionality. The unbounded case requires two separate results.

1. *Domain and self-adjointness.* If $H_0$ is self-adjoint on $D(H_0)$ and symmetric $V$ satisfies $D(H_0) \subseteq D(V)$ and is $H_0$-bounded, $\|V\psi\| \leq a\|H_0\psi\| + b\|\psi\|$, then Kato–Rellich says $H_0 + \varepsilon V$ is self-adjoint on the common domain $D(H_0)$ whenever $|\varepsilon|a < 1$. This theorem controls the operator and its domain; it does not by itself produce analytic eigenbranches or Rayleigh–Schrödinger coefficients.
2. *Analytic spectral branches.* For an analytic family with common domain and an isolated finite-multiplicity spectral cluster, analytic Riesz projections and Rellich–Kato selection give local analytic branches under the relevant hypotheses. Simple branches use Proposition 15.10; degenerate clusters first require the compression/effective-operator analysis of Theorem 15.16.

Three further cautions matter.

- *The sum may include a non-atomic integral.* The reduced resolvent is the excluded-eigenvalue integral of Proposition 15.19. A non-atomic spectral-measure component cannot be reduced to a discrete eigenprojection sum. Operator-theoretic continuous spectrum and a non-atomic spectral-measure component remain distinct notions, as in Lecture 14.
- *The bound may be unavailable.* Many important perturbations are unbounded, hence are not elements of $\mathcal{B}(\mathcal{H})$ and have no finite everywhere-defined operator norm. The disk estimate proved above then says nothing, and relative boundedness must replace it.
- *An eigenvalue can dissolve.* Under suitable coupling hypotheses, an eigenvalue embedded in continuous spectrum may cease to be an eigenvalue and be described instead by a resonance. Fermi's golden rule is a useful heuristic signpost for this mechanism, not a universal theorem that every embedded eigenvalue acquires a lifetime. A bounded reduced inverse is unavailable at such a non-isolated point.

**Example 15.21 (Optional extension: a divergent series that works).** The quartic anharmonic oscillator $H(\varepsilon) = \frac{1}{2}(\hat{p}^2 + \hat{x}^2) + \varepsilon\hat{x}^4$ has a Rayleigh–Schrödinger series with *zero* radius of convergence, yet its first few terms are numerically excellent. The reason is visible without any computation only at the level of physical motivation: for $\varepsilon < 0$ the formal potential tends to $-\infty$, signalling instability and motivating a resonant analytic continuation. But a differential expression without its domain and boundary data does not specify a self-adjoint operator, and unboundedness below implies no ground state rather than the categorical absence of all discrete spectrum. The rigorous conclusion used here is the large-order result of C. M. Bender and T. T. Wu, *Anharmonic Oscillator*, Physical Review **184** (1969), 1231–1260, doi:10.1103/PhysRev.184.1231: the perturbation series diverges and complex branch points accumulate at the origin, so its convergence radius is zero. It is nevertheless asymptotic and useful at low order. Thus "perturbation theory works" and "the perturbation series converges" are different claims. ◁

## Summary of main concepts

The recommended route within this optional capstone is finite-dimensional: coupling relative to a spectral gap controls perturbation theory. Resolvent bounds first keep the spectrum nearby; for an isolated simple level, Rayleigh–Schrödinger theory gives energy and state corrections while Hellmann–Feynman gives the exact branch derivative. The spectrum of $PVP$ gives degenerate first-order slopes, with repeated slopes leaving residual degeneracy. The further infinite-dimensional outlook explains how boundedness, isolation, and convergence can fail.

- **Operator norm and resolvent**—the norm is defined by a supremum; for finite normal $T$, $\|T\| = \max_k |\lambda_k|$ and $\|(T - zI)^{-1}\| = 1/\mathrm{dist}(z, \operatorname{spec} T)$.
- **Spectral stability**—$\operatorname{spec}(H_0 + \varepsilon V)$ stays within $|\varepsilon|\|V\|$ of $\operatorname{spec}(H_0)$.
- **Non-degenerate perturbation**—$E_n^{(1)} = \langle n| V |n\rangle$ and $E_n^{(2)} = \sum_{m\neq n} |\langle m| V |n\rangle|^2/(E_n - E_m)$.
- **State mixing**—coupling divided by gap is the coefficient of the first-order correction; the actual leading admixture magnitude also carries the factor $|\varepsilon|$.
- **Degenerate perturbation**—the spectrum of $PVP$ gives the first-order slopes with multiplicity; only distinct slopes split the level and select branch states up to phase.
- **Extension/outlook: infinite-dimensional caution**—reduced resolvents require isolation; Kato–Rellich controls self-adjointness and domains, while analytic-family hypotheses control branches; convergence is separate.